

Juneha Hwang

✉ juneha@kaist.ac.kr

🌐 <https://www.linkedin.com/in/juneha-hwang/>

🐙 <https://github.com/juneharold>

Education

Korea Advanced Institute of Science and Technology (KAIST)

August 2023-

Bachelor of Science in Computer Science (GPA: 4.01 / 4.30)

Daejeon, South Korea

- **Relevant Coursework:** Linear Algebra for Data Science (Python), Programming Principles (Rust), Data Structures (Java), Introduction to Algorithms, Machine Learning, Systems Programming (C), Programming Languages (Scala)

Experience

Linq Alpha

Jan 2025–Present

Data Engineer

Seoul, South Korea

- Engineered an entity resolution pipeline that maps ambiguous financial entity names to stock identifiers using a multi-stage agentic architecture (cache → trigram DB search → vector DB retrieval → LLM-based decision), combining RAG and structured reasoning for reliable production use.
- Built a semantic document chunking system that transforms unstructured financial documents (earnings calls, broker research, investment notes) into structured, queryable metadata using a multi-batch LLM pipeline—including multi-layer JSON parsing and validation to handle non-deterministic model outputs at scale.
- Designed and deployed a FastAPI-based API layer with asynchronous processing (Celery/Redis) for long-running LLM and data jobs, with observability instrumentation to monitor agent reliability in production.
- Managed large-scale financial datasets using PostgreSQL and Alembic, providing the retrieval backend for downstream AI pipelines.

FBA Quant (Quantitative Finance Research Group)

Feb 2024–Present

Quant Developer Track

Seoul, South Korea

- Developed expertise in asset pricing, portfolio management, and foundational financial engineering.

Projects

DEX-DEX Perpetual Futures Arbitrage Bot | Rust, Tokio, WebSockets

- Built a low-latency arbitrage bot in Rust that detects and executes spread opportunities between two perpetual futures exchanges across 96 asset pairs in real time.
- Designed a dual-channel architecture using lock-free crossbeam channels for the hot path and Tokio broadcast for cold-path recording, with a dedicated OS thread for the signal engine to avoid async runtime contention.
- Implemented a maker-race execution strategy with simultaneous post-only orders on both exchanges, sub-second fill polling, and automatic taker IOC on the opposite leg, including retry logic and reduce-only cleanup for failed fills.
- Optimized for cache performance with 64-byte aligned price state structs and atomic flags for position tracking, eliminating mutex contention on the critical path.

FABRIC Interpreter & Type Checker | Scala

- Implemented a Scala interpreter + static type checker for FABRIC, a functional language with polymorphic ADTs and pattern matching.

Certificates and Honors

Certificates

- Machine Learning with Python, Deep Learning with Keras, Deep Learning with Tensorflow (IBM)
- Using Python for Research (HarvardX)

Honors

- Samsung Collegiate Programming Cup (SCPC) 2024, 2025 Finalist
- Union of Clubs for Programming Contests (UCPC) 2024, 2025 Finalist
- Jane Street ETC 2024 1st place
- Fall 2023 Dean's List (KAIST)

Technical Skills

Programming Languages: Python, C / C++, Rust, Scala, JavaScript, Typescript, Ruby, SQL, HTML/CSS

Tech Stack: FastAPI, PostgreSQL, Celery, Redis, React, Scikit-Learn, Numpy, Pandas, Git

Languages: Korean (native), English (native), Japanese (conversational)